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1  // =====
2  //   Ver   :| Author           :| Mod. Date :| Changes Made:
3  //   V1.1 :| Alexandra Du      :| 06/01/2016:| Added Verilog file
4  // =====
5
6
7  //=====
8  //   This code is generated by Terasic System Builder
9  //=====
10
11 `define ENABLE_ADC_CLOCK
12 `define ENABLE_CLOCK1
13 `define ENABLE_CLOCK2
14 `define ENABLE_SDRAM
15 `define ENABLE_HEX0
16 `define ENABLE_HEX1
17 `define ENABLE_HEX2
18 `define ENABLE_HEX3
19 `define ENABLE_HEX4
20 `define ENABLE_HEX5
21 `define ENABLE_KEY
22 `define ENABLE_LED
23 `define ENABLE_SW
24 `define ENABLE_VGA
25 `define ENABLE_ACCELEROMETER
26 `define ENABLE_ARDUINO
27 `define ENABLE_GPIO
28
29 module DE10_LITE_Golden_Top(
30
31     /////////////// ADC CLOCK: 3.3-V LVTTL ///////////////
32     `ifdef ENABLE_ADC_CLOCK
33         input                ADC_CLK_10,
34     `endif
35     /////////////// CLOCK 1: 3.3-V LVTTL ///////////////
36     `ifdef ENABLE_CLOCK1
37         input                MAX10_CLK1_50,
38     `endif
39     /////////////// CLOCK 2: 3.3-V LVTTL ///////////////
40     `ifdef ENABLE_CLOCK2
41         input                MAX10_CLK2_50,
42     `endif
43
44     /////////////// SDRAM: 3.3-V LVTTL ///////////////
45     `ifdef ENABLE_SDRAM
46         output                [12:0]    DRAM_ADDR,
47         output                [1:0]    DRAM_BA,
48         output                DRAM_CAS_N,
49         output                DRAM_CKE,
50         output                DRAM_CLK,
51         output                DRAM_CS_N,
52         inout                 [15:0]   DRAM_DQ,
53         output                DRAM_LDQM,
54         output                DRAM_RAS_N,
55         output                DRAM_UDQM,
56         output                DRAM_WE_N,
57     `endif
```

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58
59 ////////////////////////////////////////////////// SEG7: 3.3-V LVTTL ///////////////////////////////////
60 `ifdef ENABLE_HEX0
61     output            [7:0]    HEX0,
62 `endif
63 `ifdef ENABLE_HEX1
64     output            [7:0]    HEX1,
65 `endif
66 `ifdef ENABLE_HEX2
67     output            [7:0]    HEX2,
68 `endif
69 `ifdef ENABLE_HEX3
70     output            [7:0]    HEX3,
71 `endif
72 `ifdef ENABLE_HEX4
73     output            [7:0]    HEX4,
74 `endif
75 `ifdef ENABLE_HEX5
76     output            [7:0]    HEX5,
77 `endif
78
79 ////////////////////////////////////////////////// KEY: 3.3 V SCHMITT TRIGGER ///////////////////////////////////
80 `ifdef ENABLE_KEY
81     input             [1:0]    KEY,
82 `endif
83
84 ////////////////////////////////////////////////// LED: 3.3-V LVTTL ///////////////////////////////////
85 `ifdef ENABLE_LED
86     output            [9:0]    LEDR,
87 `endif
88
89 ////////////////////////////////////////////////// SW: 3.3-V LVTTL ///////////////////////////////////
90 `ifdef ENABLE_SW
91     input             [9:0]    SW,
92 `endif
93
94 ////////////////////////////////////////////////// VGA: 3.3-V LVTTL ///////////////////////////////////
95 `ifdef ENABLE_VGA
96     output            [3:0]    VGA_B,
97     output            [3:0]    VGA_G,
98     output            VGA_HS,
99     output            [3:0]    VGA_R,
100    output            VGA_VS,
101 `endif
102
103 ////////////////////////////////////////////////// Accelerometer: 3.3-V LVTTL ///////////////////////////////////
104 `ifdef ENABLE_ACCELEROMETER
105     output            GSENSOR_CS_N,
106     input             [2:1]    GSENSOR_INT,
107     output            GSENSOR_SCLK,
108     inout             GSENSOR_SDI,
109     inout             GSENSOR_SDO,
110 `endif
111
112 ////////////////////////////////////////////////// Arduino: 3.3-V LVTTL ///////////////////////////////////
113 `ifdef ENABLE_ARDUINO
114     inout             [15:0]    ARDUINO_IO,
```

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115     inout                ARDUINO_RESET_N,
116 `endif
117
118     /////////////// GPIO, GPIO connect to GPIO Default: 3.3-V LVTTL ///////////////
119 `ifdef ENABLE_GPIO
120     inout                [35:0]        GPIO
121 `endif
122 );
123
124
125 wire [2:0]              column;
126 wire                   row;
127
128 seq_counter #(
129     .CLK_FREQ            (50000000),
130     .UPDATE_HZ           (2)
131 ) sequence_counter (
132     .clk_i                (MAX10_CLK1_50),
133     .rst_ni               (KEY[0]),
134     .column_o             (column),
135     .row_o                (row)
136 );
137
138 display display_output (
139     .clk_i                (MAX10_CLK1_50),
140     .rst_ni               (KEY[0]),
141     .column_i             (column),
142     .row_i                (row),
143     .HEX0                 (HEX0),
144     .HEX1                 (HEX1),
145     .HEX2                 (HEX2),
146     .HEX3                 (HEX3),
147     .HEX4                 (HEX4),
148     .HEX5                 (HEX5)
149 );
150 endmodule
151
```